NHH

TERM PAPER
FIE463

Spring, 2026

Start: Thursday, March 12, 09:00

End: Wednesday, March 18 12:00

THE TERM PAPER SHOULD BE SUBMITTED IN WISEFLOW AND CANVAS

You can find information on how to submit your paper here:
https://www.nhh.no/en/for-students/examinations/home-exams-and-assignments/

Your candidate number will be announced on StudentWeb. The candidate number should be noted on all pages (not your name or student number). In case of group examinations, the candidate numbers of all group members should be noted.

Collaboration between individuals or groups on submission preparation, as well as exchange of self-produced materials between individuals or groups is prohibited. The answer paper must consist of individual's or the group's own assessments and analysis. All communication during the home exam is considered cheating. All submitted assignments are processed in Ouriginal, a plagiarism control system used by NHH. Use of artificial intelligence (AI) in this exam is determined by the course coordinator. Information on AI usage and guidelines can be found on the course's Canvas page.

SUPPLEMENTARY REGULATIONS FOR EXAMINATIONS

You can find supplementary regulations under the headline “Regulations”

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Find more information under chapter 4.0 in the Supplementary provisions to the regulations for fulltime study programmes

Number of pages, including front page: 10

Number of attachments: 0

Group term paper 1

FIE463: Numerical Methods in Macroeconomics and Finance using Python

About the group term paper

- Deadline: Wednesday, March 18 at 12:00 (noon). Late submission will not be accepted.
- Your solution needs to be submitted **both on Wiseflow** (for grading) **and Canvas** (for the peer review).
- Submissions must be made in groups of 2–3 students. Cooperation across groups is not allowed.
- You are allowed to use AI for support, but you are not allowed to hand in output directly produced by AI as your own solution (see AI Regulations on the Canvas course page).

Requirements

- You must submit a *single* Jupyter notebook which contains your solution.
- Make sure your notebook runs without errors (`Restart` and `Run all`).
- Your implementation *must* work with the *FIE463* Anaconda environment we created in the course.
- Your notebook should be well structured and visually appealing, e.g.,:
  - Use markdown cells with headings and other formatting elements (see here and here).
  - Each part should be in a clearly distinguishable subsection.
  - Each function should be defined in its own code cell.
  - Each graph should be in its own cell.
  - Graphs should be visually appealing. They should include legends, labels, and other annotations so they are easy to comprehend.
  - Don't print hundreds of lines of output in your final notebook.
- Your code should satisfy the following criteria:
  - Write code that is reusable and avoid duplication. An excellent submission should contain only one implementation of each function that is called repeatedly to solve different tasks.
  - Your code should be efficient. For example, use NumPy arrays and vectorization instead of loops where it makes sense.
  - Your code should be commented to help the reader understand what it is doing.
  - Your code should be well formatted. You can use the Ruff formatter VS Code extension for this purpose.
- Be sure to provide intuitive explanations for your results when asked to do so. These explanations should be *concise*, not more than 200 words each. Expalanations *must* be provided as markdown text, *not* as Python code comments.

Assessment

Your term paper will be assessed according to the following criteria:

Component	Points
Solution for Part 1	10
Solution for Part 2	15
Solution for Part 3	20
Solution for Part 4	15
Solution for Part 5	15
Solution for Part 6	20
Notebook runs without errors	5

Note that a correct result is not sufficient to obtain full points: code efficiency, code reuse, elegance of the solution, and the presentation of the results are also factored in.

Hints

- If you are stuck, make sure to consult the lectures and workshops which cover simpler variants of the tasks below (in particular weeks 6 and 7).
- There is no need to use fancy NumPy functions we haven't covered to solve this assignment.

Overlapping generations model with workers and capitalists

Description of the economy

Recall the overlapping-generations model we studied in lecture 7. In this term paper, you are asked to extend this model along several dimensions and examine how these additions affect the equilibrium.

Households

The economy is populated by four types of households: young and old *workers*, and young and old *capitalists* (or entrepreneurs). For simplicity, we normalize the total population size to $N = 1$, and we assume that 80% of the population are workers ($N_w = 0.8$) and the remaining 20% are capitalists ($N_k = 0.2$). Each cohort survives to old age with certainty so that both workers and capitalists are equally split across the young and old cohorts (so there are $\frac{1}{2}N_w$ young and $\frac{1}{2}N_w$ old workers, and so on).

Workers

A worker household solves the following problem:

$$\begin{aligned} U_t^w = \max_{c_{y,t}^w,\, c_{o,t+1}^w} \quad & \left\{ u\left(c_{y,t}^w\right) + \beta u\left(c_{o,t+1}^w\right) \right\} \\ \text{s.t.} \quad c_{y,t}^w &= (1-\tau) w_t h_t \\ c_{o,t+1}^w &= p_{t+1} \end{aligned} \tag{1}$$

where w_t is the wage rate, τ is a payroll tax on earnings, and p_{t+1} is the pension workers receive in retirement, all of which are taken as given by worker households. Workers choose consumption $c_{y,t}^w$ when young, and consumption $c_{o,t+1}^w$ when old.

Per-period utility $u(c)$ is the CRRA utility function given by

$$u(c) = \begin{cases} \frac{c^{1-\gamma}}{1-\gamma} & \text{if } \gamma \neq 1 \\ \log(c) & \text{if } \gamma = 1 \end{cases} \tag{2}$$

where γ is the RRA coefficient and $\log(\bullet)$ denotes the natural logarithm.

For simplicity, we assume that young workers supply one unit of labor inelastically, so $h_t = 1$. It immediately follows that for given h_t, the consumption choices are determined by the budget constraints and thus no optimization is needed for the worker household.

Capitalists

Capitalists are born with initial wealth b_t which they inherit from the old capitalist generation. They do not work, but decide how much to save in assets a_t. In old age, they live off their capital returns and leave the amount b_{t+1} as an inheritance to the young capitalists.

Their maximization problem is given by

$$\begin{aligned} U_t^k = \max_{c_{y,t}^k,\, c_{o,t+1}^k,\, a_t} \quad & \left\{ u\left(c_{y,t}^k\right) + \beta u\left(c_{o,t+1}^k\right) \right\} \\ \text{s.t.} \quad & c_{y,t}^k + a_t = b_t \\ & c_{o,t+1}^k = (1 + r_{t+1})a_t - b_{t+1} \\ & c_{y,t}^k \geq 0, \;\; c_{o,t+1}^k \geq 0, \;\; a_t \geq 0 \qquad \forall\, t \end{aligned} \tag{3}$$

Unlike workers, capitalists don't receive a state pension since they never worked. The utility function $u(\bullet)$ is the same as the one for workers given in (2).

We assume that the old capitalists leave as a bequest a fixed fraction φ of their wealth when old, so that bequests are given by $b_{t+1} = \varphi(1 + r_{t+1})a_t$, and they consume the remainder $c_{o,t+1}^k = (1 - \varphi)(1 + r_{t+1})a_t$.

Firms

The firm's problem is identical to what we studied in lectures 6 and 7: A single representative firm combines capital K and labor L in a Cobb-Douglas production function to produce output Y,

$$Y_t = zK_t^\alpha L_t^{1-\alpha}$$

and maximizes profits Π according to

$$\max_{K_t,\, L_t} \;\; \Pi_t = zK_t^\alpha L_t^{1-\alpha} - (r_t + \delta)K_t - w_t L_t$$

where $(r_t + \delta)K_t$ is the cost of capital and $w_t L_t$ is the cost of labor. We assume that capital depreciates by a fraction δ each period so that r_t is the net return on capital after depreciation.

Government

The government runs a social security system with a balanced budget, such that in each period t the payroll taxes collected from young workers equal the pensions paid to old workers, $p_t = \tau w_t$.

Equilibrium

The general equilibrium in this economy is a set of quantities $(K_t, L_t, Y_t, c^w_{y,t}, c^w_{o,t}, c^k_{y,t}, c^k_{o,t}, a_t, b_t, p_t)$ and prices (r_t, w_t) which solve the households' and firm's problems such that the following market clearing conditions are satisfied:

- Asset market: $K_t = \frac{1}{2} N_k \cdot a_{t-1}$
- Labor market: $L_t = \frac{1}{2} N_w$
- Goods market: $Y_t + (1-\delta)K_t = \frac{1}{2} N_w \left(c^w_{y,t} + c^w_{o,t}\right) + \frac{1}{2} N_k \left(c^k_{y,t} + c^k_{o,t} + a_t\right)$

Additionally, the following holds:

- Social security system budget balance: $p_t = \tau w_t$
- Intergenerational transfers among capitalists: $b_t = \varphi(1+r_t)a_{t-1}$

Analytical results

In the lecture we derived the Euler equation for this type of problem. Since only the capitalists have an intertemporal problem, their Euler equation is given by

$$u'\left(c^k_{y,t}\right) = \beta(1+r_{t+1})(1-\varphi)u'\left(c^k_{o,t+1}\right)$$

Plugging in the per-period budget constraints, the capitalists' Euler equation can be expressed as a function of a single choice variable a_t,

$$u'\big(b_t - a_t\big) = \beta(1+r_{t+1})(1-\varphi)u'\big((1-\varphi)(1+r_{t+1})a_t\big) \tag{4}$$

In equilibrium, the young capitalist household chooses a_t so that this equation is satisfied.

Note that the right-hand side of the Euler equation contains the additional term $(1-\varphi)$ since the capitalists internalize that they will transfer a fraction φ to the next generation, and hence marginal changes in saving do not fully translate into next-period consumption.

Parameters

Assume that one period corresponds to 30 years and use the following parameterization:

Parameter	Description	Value
β	Discount factor	0.99^{30}
γ	Relative risk aversion	2.1
τ	Payroll tax rate	0.3
φ	Fraction of wealth bequeathed by old capitalists	0.5
α	Capital share in production	0.36
δ	Capital depreciation	$1 - 0.94^{30}$
z	Total factor productivity (TFP)	1.0
N_w	Fraction of workers	0.8
N_k	Fraction of capitalists	0.2

Note: Because of the 30-year periods, the discount factor is computed as the standard annual discount factor taken to the power of 30. Similarly, the depreciation rate is obtained by applying an annual depreciation rate of 6% over 30 years.

Part 1 — Euler equation errors

To find the equilibrium both in steady state and along a transition path, you first need to implement a function to evaluate the capitalists' Euler equation errors.

1. Define a class `Parameters` to store the model parameters.
2. Implement a function `euler_err()` with the following signature:

   ```
   def euler_err(K_next, K, par: Parameters):
       """
       Compute the capitalists' Euler equation error for a given capital
       stock today and tomorrow.
       """
   ```

   where `K_next` is the capital stock next period K_{t+1}, `K` is the current capital stock K_t and `par` are the model parameters. This function should return the Euler equation error (the left-hand side minus the right-hand side) of (4).

   Hint: The Euler equation in (4) is expressed in terms of household-level variables b_t, a_t, etc. Incorporate the market-clearing conditions to restate the equation in terms of aggregate capital.
3. Plot the Euler equation errors for a grid of capital levels K to verify that this function has a root, assuming that the capital stock is the same today and tomorrow, i.e., $K_{t+1} = K_t$. Choose a suitable interval for K to *clearly* show that this function has a root.

Part 2 — Steady state

In this part, you are asked to compute the steady state equilibrium of the economy, i.e., the equilibrium in which all quantities and prices are constant over time:

1. Define a class `SteadyState` that stores all steady-state quantities and prices.
2. Implement a function `compute_steady_state()` which takes the model parameters as an argument and computes and returns the steady state. Use a root-finder and the `euler_err()` you implemented above for this purpose.

   Hint: By definition, in steady state we have $K_{t+1} = K_t = K^*$, so that the Euler equation errors have to be zero when evaluating `euler_err(K, K, par)` at the steady-state capital stock K^*.

   Hint: Use the goods market clearing condition to verify that the equilibrium you found is correct.
3. Compute and report the steady state equilibrium for this model (this report should at least include output, capital, factor prices, and the choices of workers and capitalists).
4. Recall that one model period corresponds to 30 years. Using the equilibrium interest rate, compute the implied *annualized* net interest rate to check whether the results make sense.

Part 3 — Transition dynamics

You are now asked to investigate how the economy behaves outside of the steady state along a transition path.

1. Create a `Simulation` class to store the simulated time series of all model variables (each time series is stored as a NumPy array).

2. Implement a function `simulate_olg()` which takes as arguments an initial capital stock `K0`, the number of periods to simulate, `T`, and the model parameters. The function should compute the transition path for all variables for the periods $t = 0, \ldots, T-1$.

   Use the function `euler_err()` you wrote earlier to find the next-period capital stock which satisfies the household's Euler equation in each period t.

   Hint: You need to use a root-finder in each period. You can directly use `euler_err(K_next, K, ...)` for this since the current capital stock K_t is predetermined and thus known in period t, so the root-finder only needs to locate K_{t+1}.

   Hint: Use the goods market clearing condition *in each period* to verify that the K_{t+1} you found is correct.

3. To verify whether you computed the steady state in part 2 correctly, simulate the economy starting from an initial capital stock $K_0 = \frac{K^*}{2}$ where K^* is the steady-state capital stock, and compute the transition path for $T = 20$ periods. The economy should converge to the steady state you found.

4. Create a figure with 5-by-2 subplots showing the simulated series for Y_t, K_t, w_t, r_t, b_t, the savings rate s_t defined as $s_t = \frac{a_t}{b_t}$, and the consumption of young and old workers, and the consumption of young and old capitalists. All time series except for r_t and s_t should be shown as percent *deviations* from their steady-state values, while r_t and s_t should be shown in percent.

   Hint: You should adapt the `plot_simulation()` function from week 7 for this purpose.

Part 4 — Introduction of a wealth tax

Imagine now that the government decides to introduce a wealth tax τ_a on the assets accumulated by the old capitalists and distributes the tax revenue as a lump-sum transfer T_a to *all* households, including the capitalists themselves.

Workers

The workers' utility function remains unchanged, but their budget constraints now read:

$$\begin{aligned} c^w_{y,t} &= (1-\tau) w_t h_t + T_{a,t} \\ c^w_{o,t+1} &= p_{t+1} + T_{a,t+1} \end{aligned}$$

As before, the workers continue to supply one unit of labor and don't optimize.

Capitalists

The capitalists' budget constraints are now given by

$$\begin{aligned} c^k_{y,t} + a_t &= b_t + T_{a,t} \\ c^k_{o,t+1} &= (1-\varphi)(1-\tau_a)(1+r_{t+1})a_t + T_{a,t+1} \end{aligned}$$

Note that bequests are determined as a fraction of *after-tax wealth* (but before transfers) so that $b_{t+1} = \varphi(1-\tau_a)(1+r_{t+1})a_t$.

Additionally, the Euler equation needs to be adapted to incorporate the wealth tax as follows:

$$u'\big(b_t + T_{a,t} - a_t\big) = \beta(1+r_{t+1})(1-\tau_a)(1-\varphi)u'\Big((1-\varphi)(1-\tau_a)(1+r_{t+1})a_t + T_{a,t+1}\Big)$$

The savings rate of young capitalists is now defined as $s_t = \frac{a_t}{b_t + T_{a,t}}$.

Government

The government runs a balanced budget so that the revenues from the wealth tax have to equal the aggregate transfers distributed to all households:

$$T_{a,t} = \frac{1}{2} N_k \cdot \tau_a (1 + r_t) a_{t-1}$$

This follows since there are $\frac{1}{2}N_k$ old capitalists paying the tax which is redistributed to the entire population with mass $N = 1$.

Perform the following steps to incorporate the wealth tax into your previous analysis:

1. Update the `Parameters` class to include the wealth tax rate and set $\tau_a = 0.05$.
2. Modify the functions you wrote earlier to implement the wealth tax. Note that this only requires a few changes in each function.
3. Compute and report the new steady state with the wealth tax. What happens to the savings rate and the capital stock in the new steady state? Briefly explain the intuition for your findings.
4. Simulate the transition path for $T = 20$ periods when the economy starts in the steady state without a wealth tax. The wealth tax is introduced as a surprise in period $t = 0$ and applied immediately to the savings accumulated from the previous period.
5. Plot the transition paths for the same variables as in part 3. Indicate the old and new steady states graphically to verify that the economy converges to the new steady state.
6. Briefly discuss who is better or worse off along the transition and in the new steady state.

Hint: Do not copy the functions and classes you wrote earlier but instead adapt them so they can be used to solve both the case of $\tau_a = 0$ (Parts 1–3) and $\tau_a > 0$ (Part 4). Running Part 4 should then require only a few lines of extra code.

Part 5 — Social welfare and the optimal wealth tax

You are interested in whether the average household in this economy benefits from the introduction of the wealth tax or not. To simplify matters, we're going to ignore the transition after the introduction of the wealth tax and simply compare the steady states with and without a wealth tax according to the utilitarian social welfare function.

The utilitarian social welfare function computes the weighted average of household utilities in the economy. For the case without a wealth tax, let this social welfare function be defined as

$$SW_{\tau_a=0} = \frac{1}{2} N_w \left(U_y^w + U_o^w \right) + \frac{1}{2} N_k \left(U_y^k + U_o^k \right)$$

where U_y^w is the utility of a young worker defined in (1), $U_o^w = u(c_o^w)$ is the utility of an old worker (who dies at the end of the period and thus the utility is just the CRRA utility function), U_y^k is the utility of a young capitalist defined in (3), and $U_o^k = u(c_o^k)$ is the utility of an old capitalist. All of these utilities should be evaluated at the steady-state values of consumption in the economy without a wealth tax.

The social welfare function for the economy *with* a wealth tax, $SW_{\tau_a>0}$ is computed analogously, but using the household choices for the steady state *with* a wealth tax.

1. Compute the social welfare for the steady states without and with the wealth tax. Which steady state would households prefer to live in, on average? Which types of households are driving these results?
2. Find the socially optimal wealth tax rate. To this end, write code to maximize social welfare as a function of τ_a, and report this optimal tax rate τ_a^*.
3. Solve for the steady state using τ_a^*, and report the percentage change in output Y and capital K vs. the original steady state without a wealth tax. Briefly explain how this new steady state can be welfare maximizing if both output and capital are lower.

Part 6 — Uncertainty about wealth taxes

Finally, consider the scenario where there is uncertainty about the wealth tax policy in the next period. This might arise because left- and right-wing parties alternate in government and they adapt the tax policy when being elected. Mathematically, assume that the wealth tax rate follows an IID process with

$$\tau_{a,t+1} = \begin{cases} 0 & \text{with probability } 1 - \pi_a \\ \tau_a^{iid} & \text{with probability } \pi_a \end{cases}$$

Assume that $\tau_a^{iid} = 0.05$ and that the probability of a wealth tax being enacted is $\pi_a = 0.5$. Because the wealth tax is IID, next period's $\tau_{a,t+1}$ does not depend on whether there is a wealth tax today or not.

Workers

Workers are uncertain about the level of transfers $T_{a,t+1}$ they receive when old, but since they don't make any choices in this model, this uncertainty will only affect their old-age consumption, as implied by their budget constraint.

Capitalists

The capitalists now maximize *expected* lifetime utility,

$$\begin{aligned} U_t^k = \max_{c_{y,t}^k,\, c_{o,t+1}^k,\, a_t} & \left\{ u\left(c_{y,t}^k\right) + \beta \mathbb{E}\left[u\left(c_{o,t+1}^k\right)\right] \right\} \\ \text{s.t.} \quad & c_{y,t}^k + a_t = b_t + T_{a,t} \\ & c_{o,t+1}^k = (1-\varphi)(1-\tau_{a,t+1})(1+r_{t+1})a_t + T_{a,t+1} \end{aligned}$$

Note that capitalists are uncertain about both the prevailing tax rate $\tau_{a,t+1}$ as well as the level of transfers $T_{a,t+1}$ next period. However, the transfers $T_{a,t}$ received when young are not uncertain since households know whether or not there is a wealth tax in place today. Bequests in period $t+1$ are also uncertain and given by $b_{t+1} = \varphi(1-\tau_{a,t+1})(1+r_{t+1})a_t$.

Moreover, the Euler equation has to be adapted to reflect the uncertainty about the tax policy next period:

$$u'\left(b_t + T_{a,t} - a_t\right) = \beta(1+r_{t+1})(1-\varphi)\mathbb{E}\left[(1-\tau_{a,t+1})u'\Big((1-\varphi)(1-\tau_{a,t+1})(1+r_{t+1})a_t + T_{a,t+1}\Big)\right]$$

1. Update the `Parameters` class to include the new attributes introduced in this part.
2. Implement a function `euler_err_uncertain()` which evaluates the Euler equation error for the updated Euler equation to take into account the uncertainty about next period's tax rate.
3. Solve for the new steady state when there is uncertainty about the wealth tax policy next period, but actual wealth tax remains at $\tau_{a,t} = 0$ forever. Is this steady state identical to the one you solved for in part 2? Why or why not?

   Hint: The only change required to `compute_steady_state()` is to use `euler_err_uncertain()` instead of `euler_err()`.
4. Adapt the `simulate_olg()` function you wrote earlier so that you can compute transition paths with uncertainty about the wealth tax policy. Starting at the original steady state from part 2, simulate the economy for $T = 20$ if future tax policy becomes uncertain in period $t = 0$, but the actual wealth tax rate remains at $\tau_{a,t} = 0$ for the entire simulation.

   Hint: The only change required to `simulate_olg()` is to use `euler_err_uncertain()` instead of `euler_err()`.
5. How does the new steady state compare to the ones you found in parts 2 and 5? Briefly explain the key differences and the underlying mechanisms.

Note: You should go back and adapt the implementations of `compute_steady_state()` and `simulate_olg()` so they can work for both the deterministic and the stochastic wealth taxes.